

502 1st

Course 502

Time 50 minutes

Marks 20

1. (a) Show the impacts of saving ratio in the steady state solution of Solow growth model and prove that exogenous changes in the saving ratio leave only level effects on k and y , but not growth effects.

(b) Show the impacts of changes in the saving ratio on per capita consumption and long run output.

Does any steady state solution automatically imply golden rule of capital accumulation? Discuss.

[6+6]

2. Find the elasticity of output per unit of effective labor on the balanced growth path, y^* , with respect to the rate of population growth, n . If $\alpha_k(k^*) = 1/3$, $g = 2\%$, and $\delta = 3\%$, by about how much does a fall in n from 2 percent to 1 percent raise y^* ? [4]

3. Briefly explain classical dichotomy and neutrality of money using the pillars of classical macroeconomics.

[4]

Answer any one

1. Consider a Ramsey-Cass-Koopmans model with the production function $Y = F(K, AL)$ and real interest rate r ; household's utility function takes the form

$$U = \int_0^{\infty} e^{-\rho t} u(c(t)) \frac{L(t)}{L} dt$$

and there is CRRA (constant -relative -risk-aversion) utility

$$U(c(t)) = \frac{c(t)^{1-\theta}}{1-\theta}$$

- Construct Household budget constraint and show that there is no-Ponzi-game condition. [5] - Pg-53-54 51-53 (73-75)
- Solve Euler equation for households' maximization problem. [7] (53-56) (75-78)
- Describe the behavior of the economy in terms of the evolution of c and k . [4] (57-59) (79-81)
- How does the competitive equilibrium maximize the welfare of the representative household. [4] (63-64) (85-86)

2. Suppose $Y_t = F(K_t, A_t L_t)$, with having constant returns to scale and the intensive form of the production function satisfying the Inada conditions. Suppose also that $A_{t+1} = (1+g)A_t$, $L_{t+1} = (1+n)L_t$ and the production function is Cobb-Douglas. Assume time is discrete rather than continuous and each individual lives for only two periods.

- Solve the individual's maximization problem where θ determines how much individuals varies in response to differences between r and ρ . [5] (78-80) (100-102)
- Find an expression for k_{t+1} as a function of k_t . [4] (81-82) (103-104)
- Describe how each of the following affects k_{t+1} as a function of k_t :
 - A rise in n .
 - A rise in α . [3]

- (d) Sketch k as a function of k_t . Does the economy have a balanced growth path? If the initial level of k differs from the value on the balanced growth path, does the economy converge to the balanced growth path? [4]

- Find the rate of convergence of k to k^* . [4] (84-85) (106-107)

3. Consider a simplified version of the models of R&D and growth developed by P. Romer (1990), Gross and Helpman (1991) and Aghion and Howitt (1992) where θ reflects the effect of the existing stock of knowledge on the success of R&D.

- Explain the simplification process to construct the framework. [2] (102-104) (124-126)
- Critically explain the role of a_L (the labor force used in the R&D sector) for the dynamics of knowledge accumulation without considering capital into the model. [6] (104-109) (126-131)
- Assume that $\theta + \beta < 1$ and $n > 0$, and that the economy is on its balanced growth path. Describe how each of the following changes affects the $g_A = 0$ and $g_K = 0$ lines and the position of the economy in (g_A, g_K) space at the moment of the change:
 - An increase in n .
 - An increase in a_K .
 - An increase in θ . [6]
- Assume that $\beta + \theta = 1$ and $n = 0$,
 - Find the value that A/K must have for g_K and g_A to be equal.
 - Find the growth rate of A and K when $g_K = g_A$.
 - How does an increase in s affect the long-run growth rate of the economy? [6]

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Exercise 3.5

Exercise 3.3

Course 502

Time 1 hour

Marks 20

Answer any one

1. Suppose firm i 's output is $Y_i(t) = K_i(t)^\alpha [A(t)L_i(t)]^{1-\alpha}$ and $A(t) = BK(t)$. Here K_i and L_i are the amounts of capital and labor used by firm i and K is the aggregate capital stock. Capital and labor earn their private marginal products. The economy is populated by infinitely lived households that own the economy's capital stock. Individual's discount rate is ρ and their instantaneous utility function is logarithmic. The utility of the representative household takes the constant-relative-risk aversion form and θ is the coefficient of the relative risk aversion. Population growth is zero.
- How the long-run growth is endogenous which is effected by the saving rate according to learning by doing? [4]
 - When output is produced by competitive firms, what will be the demand for the holder of the patent on an idea and the profit? [4]
 - Describe the assumptions to construct the model. [4]
 - When the R&D will be zero and how the long-run growth will be determined? [4]
 - What will be the socially optimal level of L_A ? Explain how it is different from the equilibrium level of L_A . [4]

2. Consider the infinitely lived representative household that derives utility from both consumption (c_t) and leisure ($1-l_t$). The production function of the representative firm is given by

$$f(k_t, l_t, z_t) = z_t k_t^\alpha l_t^{1-\alpha}$$

$$z_{t+1} = \phi z_t + \varepsilon_{t+1}$$

where z_t is the productivity parameter ε is a white noise error term.

- Construct the household's problem and using log utility derive the short-run labor supply curve. Explain the relationship among leisure, real wage, interest rate. [3]
- Prove that long run labor supply is independent of wage when $\phi = \frac{1}{3}$ [3]
- Assuming $\frac{1+r}{1+\rho} = 1$ and functional form of utility of leisure is $v(h) = \frac{\sigma}{1-\sigma} (h)^{\frac{\sigma-1}{\sigma}}$, find the elasticity of labor supply. [4]
- Mention two important limitations of previous business cycle models. Discuss Hansen's indivisible labor supply solution as an attempt to overcome these limitations. [6]
- What are the stylized facts of real business cycle? [4]

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